

1.5: Series

1. Summation Notation

$\sum_{r=a}^b f(r) = f(a) + f(a+1) + \dots + f(b)$. Summation is linear:

$$\sum [\lambda f(r) + \mu g(r)] = \lambda \sum f(r) + \mu \sum g(r)$$

2. Standard Sums

$$\sum_{r=1}^n 1 = n, \quad \sum_{r=1}^n r = \frac{n(n+1)}{2}$$

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Find $\sum_{r=1}^n (2r^2 - 3r + 1)$.

$$= \frac{n(n+1)(2n+1)}{3} - \frac{3n(n+1)}{2} + n = \frac{n(4n^2 - 3n - 1)}{6}$$

Shifting Limits

$$\sum_{r=a}^b f(r) = \sum_{r=1}^b f(r) - \sum_{r=1}^{a-1} f(r)$$

3. Method of Differences

If $f(r) = g(r+1) - g(r)$, then $\sum_{r=1}^n f(r) = g(n+1) - g(1)$ (telescoping).

Show $\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$. **Hence** $\sum_{r=1}^n \frac{1}{r(r+1)}$.

Partial fractions give the identity. Telescoping: $1 - \frac{1}{n+1} = \frac{n}{n+1}$.

Using an Identity Provided

Exam questions often give a useful identity like $r(r+1)(r+2) - (r-1)r(r+1) = 3r(r+1)$, and ask you to use it to find $\sum r(r+1)$.

4. Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

Standard Series

Function	Expansion	Valid for
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	All x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	All x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	All x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$	$ x < 1$ if $n \notin \mathbb{N}$

Building New Series

- **Substitute:** e^{-2x} from e^x by replacing $x \rightarrow -2x$.
- **Multiply:** truncate and multiply products (e.g. $e^x \cos x$).
- **Differentiate / integrate** term-by-term within the radius.

5. Taylor Series

Expansion of f about $x = a$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

6. Applications

Approximations

Small- x : $\sin x \approx x$; $\cos x \approx 1 - \frac{x^2}{2}$; $e^x \approx 1 + x$.

Limits

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \text{ by substituting series.}$$

Exam Tip: Note convergence explicitly (especially for $\ln$, $(1 + x)^n$ with non-integer n). When building products, keep only the terms up to the requested order.